Chi squared test for independence

1 For each of the following matrices being examined for independence using a χ^2 test, state the null hypothesis and the number of degrees of freedom.

a

	Test score 1	Test score 2	Test score 3
Male	59	71	67
Female	86	90	89

b

	Male	Female
Teacher	241	271
Nurse	46	390
Tradie	252	52
Lawyer	154	158

c

	Junior	Senior	Veteran
Tennis	241	151	47
Rowing	76	54	32
Golf	45	254	232
Bowls	8	25	355
Swimming	155	97	167
Running	188	102	37

2 At the last Commonwealth Games, 220 sp[redacted]rs were asked which of four events they preferred to watch. Results are displayed in the following table:

	Age under 40	Age 40 or more
Hockey	35	31
Lawn bowls	6	24
Swimming	45	49
Archery	9	21

In order to see whether age had any influence over favourite sport, researchers conducted a χ^2 test of independence at a 5% significance level.

a State the hypotheses set for the problem.

b State the number of degrees of freedom.

c Calculate the test statistic, χ^2, correct to two decimal places.

d Find the p-value, correct to five decimal places.

e Comment on your findings.

3 The following table of data was collected from a survey where people were asked what their favourite movie genre was:

	Action	Comedy	Romance	Total
Male	51	50	11	112
Female	34	42	54	130
Total	85	92	65	242

Researchers conduct a χ^2 test of independence at a 1% significance level.

a State the hypotheses set for the problem.

b State the number of degrees of freedom.

c Calculate the test statistic, χ^2, correct to two decimal places.

d Given that the critical value for a 1% significance level is 11.34, state whether the null hypothesis should be accepted. Explain your answer.

4 A cleaning company is interested in determining if good results in window cleaning are dependent on the brand of cleaning product, or some other variable such as cleaning technique. They asked their clients to complete a survey after each clean and results are recorded in the table below:

	Product A	Product B	Product C	Total
Poor	6	7	3	16
Average	10	11	13	34
Good	22	21	25	68
Excellent	12	11	12	35
Total	50	50	53	153

The company conducts a χ^2 test of independence at a 5% significance level.

a State the hypotheses set for the problem.

b State the number of degrees of freedom.

c Calculate the test statistic, χ^2, correct to two decimal places.

d Find the p-value, correct to two decimal places.

e State whether the null hypothesis should be accepted. Explain your answer.

5 A generic smart phone retailer is interested in knowing more about their customer base. They conduct a survey asking customers how important the latest camera technology is to them when choosing their phone. Results are recorded in the table below:

	Under 18	$18- < 25$	$25-40$	Over 40	Total
Not important	23	27	11	32	93
Slightly important	21	18	23	54	116
Important	10	21	25	22	78
Very important	16	43	55	18	132
Total	70	109	114	126	419

The company conducts a χ^2 test of independence at a 5% significance level.

a State the hypotheses set for the problem.

b State the number of degrees of freedom.

c Find the expected frequency for people over 40 who think camera technology is very important, rounded to the nearest whole.

d Calculate the test statistic, χ^2, correct to two decimal places.

e If the critical value for this level of significance is 16.92, state whether the null hypothesis should be accepted. Explain your answer.

6 The following table shows the results of s[redacted]ng a number of adults on their number of minutes of exercise per day and measured blood pressure (bp) level:

	< 30	$30 - 60$	> 60	Total
Low bp	5	12	32	49
Average bp	7	31	6	44
High bp	22	13	5	40
Total	34	56	43	133

The company conducts a χ^2 test of independence at a 5% significance level.

a State the hypotheses set for the problem.

b Use a calculator or otherwise, to construct a table of expected frequencies for the data.

c State the number of degrees of freedom.

d Calculate the test statistic, χ^2, correct to two decimal places.

e Find the p-value, correct to two decimal places.

f State whether the null hypothesis should be accepted. Explain your answer.

7 1000 people were questioned as they exited a polling booth at the last election. The results are displayed in the following table:

	Party A	Party B	Party C	Total
Female	200	150	50	400
Male	250	300	50	600
Total	450	450	100	1000

Perform a χ^2 test for independence at a 5% significance level to determine whether voting preference of the men seems to be significantly different to that of the women.
When conducting your test be sure to state the following:

- Hypotheses and significance level.
- df, χ^2 and p-value.
- A comment on whether the hypothesis is rejected or not rejected.

8 A electronics store is interested in knowin[illegible]re about their customer base. They conduct a survey asking customers how old they were when they bought their first mobile phone and whether that phone came with internet access. Results are recorded in the table below:

	Under 8	$8- < 12$	$12-18$	Over 18	Total
Internet access	1	4	46	86	137
No internet access	13	35	7	5	60
Total	14	39	53	91	197

The company wishes to conduct a χ^2 test of independence at a 1% significance level to see if there appears to be a link between phone internet access and age.

a State the hypotheses set for the problem.

b There is less than 5 observations in the first two columns. Combine the columns together to create a new table of values.

c State the number of degrees of freedom.

d Calculate the test statistic, χ^2, correct to two decimal places.

e If the critical value for this level of significance is 9.21, state whether the null hypothesis should be accepted. Explain your answer.